Vimal Jyothi Engineering College
Department of Artificial Intelligence and Data Science

ADD415 PROJECT PHASE

CareerPrep AI:XGBoost-Powered predictive Career Matching System integrating Machine Learning and Natural Language Processing

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Objective

- A unified platform for career recommendations ensuring user satisfaction
- Match candidates with jobs and courses shortlisted using course or job recommendation model efficiently
- Improve resumes and profiles using cv optimizer.
- Offer real-time salary and market trend understanding.
- Help users prepare with AI-driven mock tests and tips.
- Ensure a smooth and scalable platform for all users.

Introduction

- Career prep AI leverages Machine Learning (XGBoost) for accurate career and job predictions, along with automated job assignment via Firebase Cloud Functions.
- This includes an integrated CV generator, resume optimizer, exam preparation tools, interview practice, and industry trend analysis.
- Also provides salary comparisons, job market insights, and a cloud-based Android app for a seamless user experience.

Literature Survey

Ref no	Paper title	Journal	Year	Authors	Description
1	AI based suit-ability measure-ment and pre-diction between job description and job seeker profiles	International Journal of Information Manage-ment Data Insights	Volume 2, 2022	Sridevi G.M , S. Kamala Suganthi	An AI system that uses text mining, NLP, and machine learning to as-sess and predict candi-date fit for job descrip-tions.
2	Enhancing personal-ized learning: AI-driven iden-tification of learning styles and content modification strategies	International Journal of Cognitive Computing in Engineer- ing	Volume 5, 2024	Md. Kabin Hasan Kanchon, Mahir Sadman, Kaniz Fatema Nabila, Ramisa Tarannum, Riasat Khan	A model that detects learning styles and tailors educational content us-ing XGBoost and ensem-ble techniques for per-sonalized learning.

Literature Survey

Contd...

Ref no	Paper title	Journal	Year	Authors	Description
3	C3-IoC:A Career Guidance System for Assessing Student Skills using Machine Learning and Network Visualisation	International Journal of Artificial Intelligence in Education	Volume 33, 2023	Ad´an Jos´e-Garc´ia, Alison Sneyd, Ana Melro, Ana¨ıs Ollagnier, Georgina Tarling	C3-IoC is an AI career guidance system that helps students explore IT paths by assessing their skills and visualizing job role networks.
4	Automatic question- answer pairs generation us- ing pre-trained large language models	Computers and Education: Artificial Intelligence	Volume 6, 2024	Jintao Ling, Muhammad Afzaal	This paper examines methods for generating question-answer pairs with pre-trained language models to improve assessment creation.

Literature Survey

Contd...

Ref no	Paper title	Journal	Year	Authors	Description
5	Spectral clustering algorithm based web mining and quadratic support vector machine for learning style prediction	Measurement: Sensors	Vol 31, 2024	K.N. Prashanth Kumar, B.T. Harish Kumar, A. Bhuvanesh	This paper proposes a new method that uses spectral clustering and quadratic SVM to predict learning styles in e-learning by analyzing web usage data.
6	Real Time Mock Interview using Deep Learning	International Journal of Engineering Research and Tech- nology (IJERT)	Vol 10, 2021	Rohan Patil, Akash Butte, Sahil Temgire, Varun Nanekar, Asst. Prof. Shivganga Gavhane	This paper describes a web app that uses deep learning to help users practice for online interviews.

Literature Survey

Contd...

Ref no	Paper title	Journal	Year	Authors	Description
7	Student Career Prediction Using Advanced Machine Learning Techniques	International Journal of Engineering and Technology	Vol 7, 2018	K. Sripath Roy, K. Roopkanth, V. Uday Teja, V. Bhavana, J. Priyanka	Focuses on predicting career paths for computer science students using advanced machine learning algorithms like SVM, Decision Tree, and XGBoost
8	Career Counselling Using Data Mining	International Journal of Innovative Research in Computer and Communication Engineering	Vol 5, 2017	Nikita Gorad, Ishani Zalte, Aishwarya Nandi, Deepali Nayak	This web application helps high school students choose career paths based on their personality traits, interests, and abilities using the C5.0 algorithm and adaptive boosting.

Literature Survey

Contd...

Ref no	Paper title	Journal	Year	Authors	Description
9	Career Choice Using Machine Learning Algorithms	2022 International Conference on Inventive Computation Technologies (ICICT)	Vol 2022, 2022	Borra Harsha, Nandeti Sravanthi, Nagella Sankeerthana, Manne Suneetha	This paper explores how Decision Trees, Random Forests, and Support Vector Machines can predict suitable career paths for students based on their skills and attributes.
10	Career Path Prediction Using Machine Learning Classification Techniques	International Journal of Scientific Research in Computer Science	Volume 8,2021	Prathamesh Gavhane, Dhanraj Shinde, Ashwini Lomte, Naveen Nattuva, Shital Mandhane	This paper presents a machine learning model that predicts suitable career paths for students based on their abilities and qualities using supervised classification techniques.

Conclusion From Literature Survey

- AI-driven systems aim to personalize career guidance and educational content based on individual skills and traits.
- Many models help students explore suitable career paths by visualizing job role networks and predicting careerfi t based on attributes.
- Advanced machine learning algorithms (e.g., XGBoost, SVM, Decision Trees) and natural language processing enhance the assessment of userfi t for job roles and learning styles.
- Tools such as mock interviews and practice tests improve users' readiness for the job market or educational challenges.

Problem Definition

- Lack of Personalized Career Guidance .
- Mismatch Between Skills and Job Market .
- Time-Consuming Job Search Process.
- Unclear Salary Industry Trends.
- Limited Resources for Interview Exam Preparation.
- Rapidly Changing Industry Demands

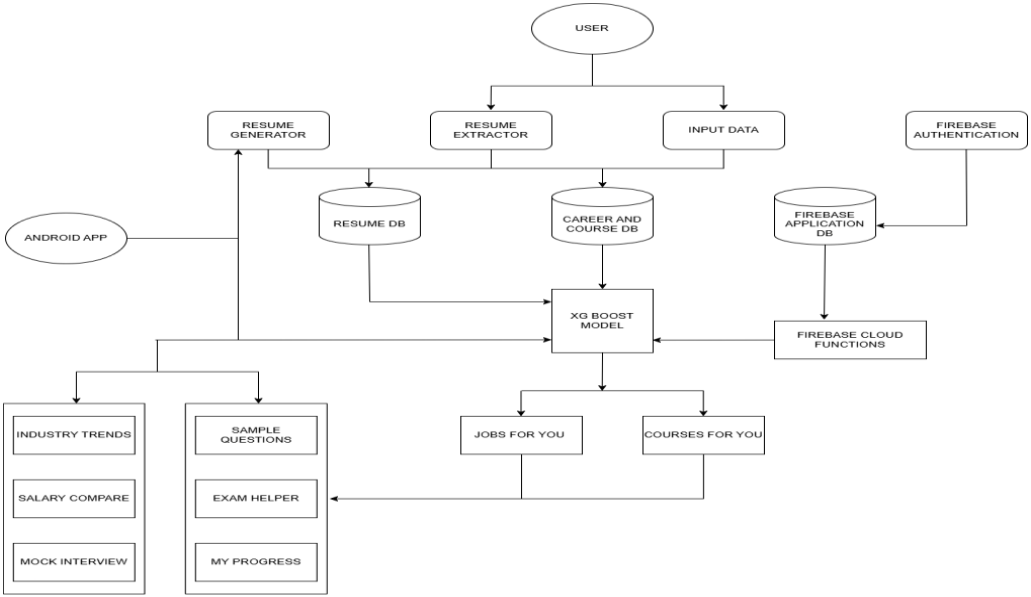


Figure: Proposed System

Methodology

- **Data Collection and Preprocessing:**Collect data related to jobs, courses, and industry trends for each of the 8 categories.Preprocess user inputs (education, skills, experience) and categorize them into the 8 areas using replace and LabelEncoder preprocessing techniques.
- **XGBoost Model for Dynamic Shortlisting:**A custom XGBoost-based machine learning model is created to analyze the user's preferences and profile data.The model dynamically shortlists jobs and courses based on the selected area of interest and client profile.

Methodology

Contd...

- **Job and Course Recommendation:** The model filters and ranks jobs and courses from relevant categories using XGBoost predictions to recommend the bestfi t for the user. Provides personalized recommendations based on both user profile and industry trends.
- **Continuous Learning and Adaptation:** The model continually learns from new data (job trends, industry changes) to refine recommendations.Adaptive system ensuring recommendations remain relevant and up-to-date.
- **Dynamic Assignment via Cloud Functions:**Job and course recommendations are automatically assigned to users using Firebase Cloud Functions, ensuring a seamless and efficient experience.

Use Case Diagram

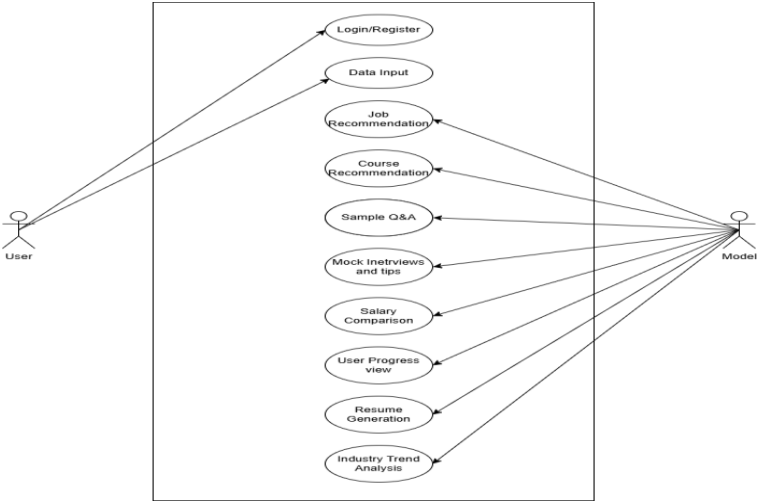
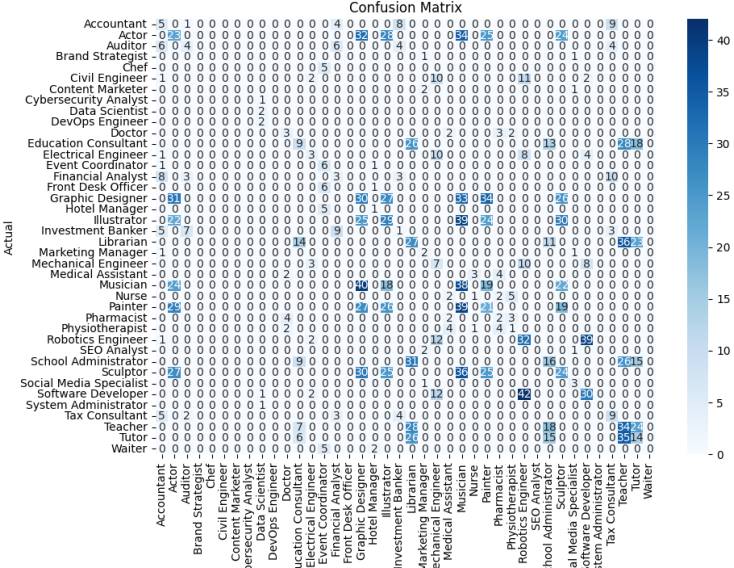


Figure: Use Case Diagram

Result


Confusion Matrix
Actual
Accountant
Actor
Auditor
Brand Strategist
Chef
Civil Engineer
Content Marketer
Cybersecurity Analyst
Data Scientist
DevOps Engineer
Doctor
Education Consultant
Electrical Engineer
Event Coordinator
Financial Analyst
Front Desk Officer
Graphic Designer
Hotel Manager
Illustrator
Investment Banker
Librarian
Marketing Manager
Mechanical Engineer
Medical Assistant
Musician
Nurse
Painter
Pharmacist
Physiotherapist
Robotics Engineer
SEO Analyst
School Administrator
Sculptor
Social Media Specialist
Software Developer
System Administrator
Tax Consultant
Teacher
Tutor
Waiter
0
5
10
15
20
25
30
35
40

Result

cont...

Classification Report:

	precision	recall	f1-score	support
Accountant	0.21	0.31	0.25	29
Actor	0.17	0.15	0.16	170
Auditor	0.24	0.15	0.18	27
Brand Strategist	0.00	0.00	0.00	2
Chef	0.42	0.62	0.50	8
Civil Engineer	0.00	0.00	0.00	24
Content Marketer	0.50	1.00	0.67	4
Cybersecurity Analyst	0.33	1.00	0.50	2
Data Scientist	0.00	0.00	0.00	2
DevOps Engineer	0.00	0.00	0.00	1
Doctor	0.24	0.42	0.30	12
Education Consultant	0.22	0.20	0.21	94
Electrical Engineer	0.22	0.33	0.26	27
Event Coordinator	0.44	0.57	0.50	7
Financial Analyst	0.50	0.20	0.29	25
Front Desk Officer	0.00	0.00	0.00	5
Graphic Designer	0.16	0.13	0.14	172
Hotel Manager	0.00	0.00	0.00	5
Illustrator	0.17	0.21	0.19	164
Investment Banker	0.21	0.48	0.29	27
Librarian	0.26	0.28	0.27	105
Marketing Manager	0.30	0.75	0.43	4
Mechanical Engineer	0.23	0.26	0.24	27
Medical Assistant	0.00	0.00	0.00	9
Musician	0.19	0.19	0.19	174
Nurse	0.15	0.15	0.15	13
Painter	0.17	0.18	0.17	165
Pharmacist	0.29	0.18	0.22	11
Physiotherapist	0.15	0.20	0.17	10
Robotics Engineer	0.39	0.28	0.33	88
SEO Analyst	0.00	0.00	0.00	5
School Administrator	0.22	0.23	0.23	103
Sculptor	0.18	0.18	0.18	167
Social Media Specialist	0.00	0.00	0.00	4
Software Developer	0.48	0.60	0.53	89
System Administrator	0.00	0.00	0.00	1
Tax Consultant	0.00	0.00	0.00	24
Teacher	0.20	0.20	0.20	96
Tutor	0.21	0.20	0.20	92
Waiter	0.10	0.17	0.12	6
accuracy			0.22	2000
macro avg	0.19	0.25	0.20	2000
weighted avg	0.21	0.22	0.21	2000

Figure: Classification Matrix

Result

cont...

```
Clients
  jyothika
    Address: "Kerala_House,_Palakkad,_Kerala-670702"
    DOB: "26-09-2004"
    Education: "B.tech_in_AI_and_DS"
    Gender: "F"
    Interests: "11100001"
    Mob: "984701097"
    Name: "Jyothika_J"
    Password: 12345678
    email: "jyothikaj@gmail.com"
```

Figure: Example Dataset structure of a client profile

Result

cont...

```
Jobs
  Illustrator
    IncraLabs
      Application: "www.incralabs.com"
      Description: "It_is_a_good_job. So_apply_now_and_enjoy"
      Location: "Banglore"
      Logo: "https://incralabs.com/wp-content/uploads/2024/07/LogoBR-300x96.png"
      Salary: "30,000"
      Title: "Adobe_Illustrator_Designer"
```

Figure: Example Database structure of a job

Result

cont...

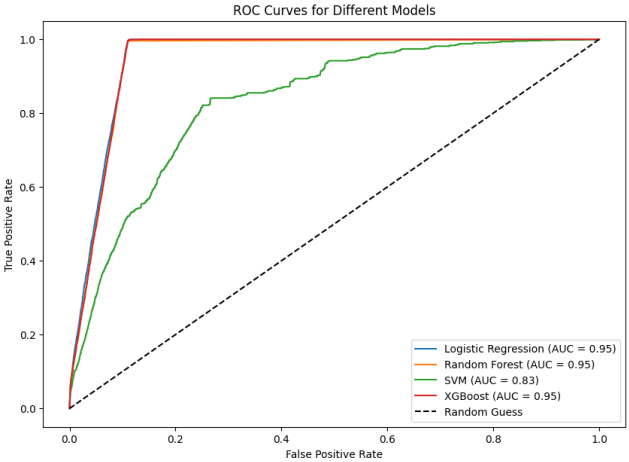
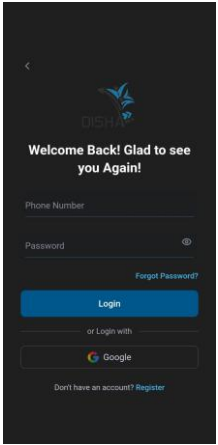


Figure: ROC Curve

Result

cont...



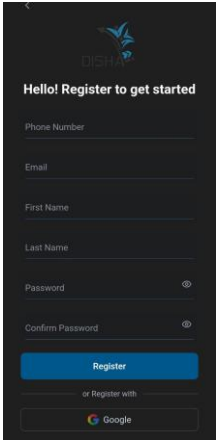
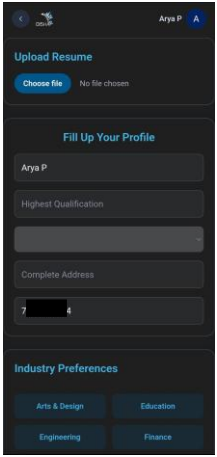


Figure: Login and Register

Result

cont...



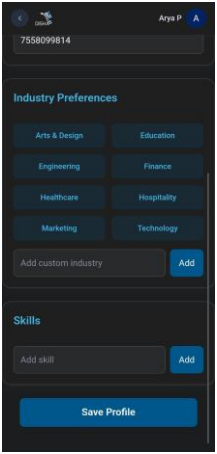


Figure: User Details

Result

cont...

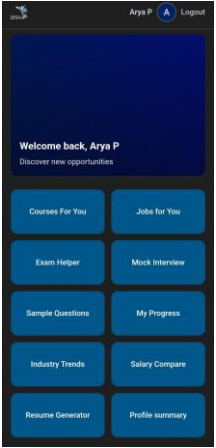
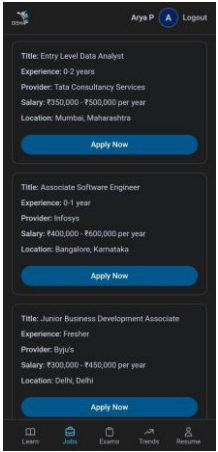


Figure: Home Screen

Result

cont...



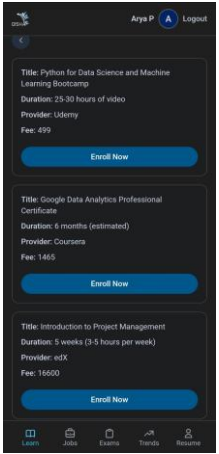
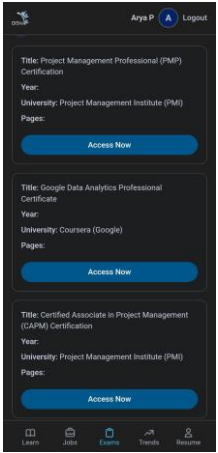


Figure: Jobs and Courses predicted

Result

cont...



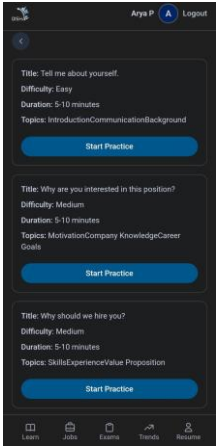
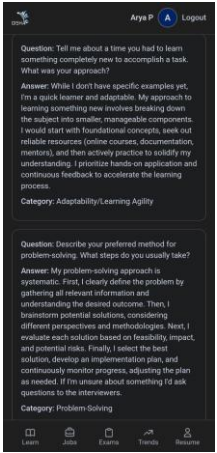


Figure: Exam Helper and Mock Interviews

Result

cont...



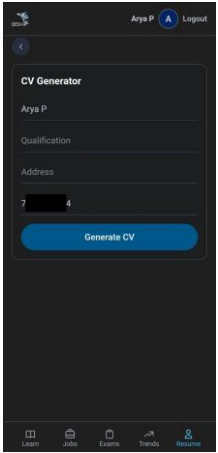
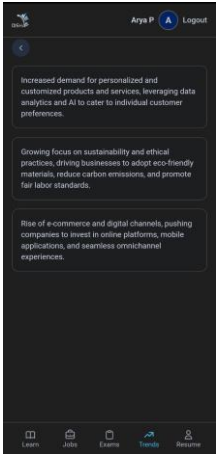


Figure: Sample questions and resume generator

Result

cont...



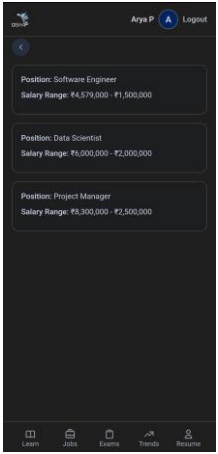


Figure: Industry Trends and Salary Comparison

Conclusion

- The project utilizes advanced machine learning algorithms, particularly XGBoost , to provide tailored career predictions based on individual skills, interests, and attributes, enhancing the decision-making process for students
- The platform - Disha AI, incorporates features such as mock interviews, aptitude tests, and personalized tips, equipping students with the necessary tools to improve their readiness for both job markets and educational challenges.
- By integrating AI-driven systems and data visualization techniques, Disha AI addresses the evolving needs of career counseling, ensuring that candidates receive relevant and effective guidance in their career exploration journey.

References

- G. Sridevi and S. K. Suganthi, “A i based suitability measurement and prediction between job description and job seeker profiles,” International Journal of Information Management Data Insights, vol. 2, no. 2, p. 100 109, 2022.
- M. K. H. Kanchon, M. Sadman, K. F. Nabila, R. Tarannum, and R. Khan, “Enhancing personalized learning: Ai-driven identification of learning styles and content modification strategies,” International Journal of Cognitive Computing in Engineering, vol. 5, pp. 269–278, 2024.
- A. Jos´e-Garc´ıa, A. Sneyd, A. Melro, A. Ollagnier, G. Tarling, H. Zhang, M. Stevenson, R. Everson, and R. Arthur, “C3-ioc: A career guidance system for assessing student skills using machine learning and network visualisation,”International Journal of Artificial Intelligence in Education, vol. 33, no. 4,
pp. 1092–1119, 2023.

References

Contd...

- J. Ling and M. Afzaal, “Automatic question-answer pairs generation using pre-trained large language models in higher education,” Computers and Education: Artificial Intelligence, p. 100 252, 2024.
- K. P. Kumar, B. H. Kumar, and A Bhuvanesh, “Spectral clustering algorithm based web mining and quadratic support vector machine for learning style prediction in e-learning platform,” Measurement: Sensors, vol. 31, p. 100 962, 2024.
- R. Patil, A. Butte, S. Temgire, V. Nanekar, and S. Gavhane, “Real time mock interview using deep learning,” IJERT, vol. 10, no. 5, 2021.

References

Contd...

- K. S. Roy, K Roopkanth, V. U. Teja, V Bhavana, and J Priyanka, “Student career prediction using advanced machine learning techniques,” International Journal of Engineering & Technology, vol. 7, no. 2.20, pp. 26–29, 2018.
- N. Gorad, I. Zalte, A. Nandi, and D. Nayak, “Career counselling using data mining,” International Journal of Innovative Research in Computer and Communication Engineering, vol. 5, no. 4, 2017.
- B. Harsha, N. Sravanthi, N. Sankeerthana, and M. Suneetha, “Career choice using machine learning algorithms,” in 2022 International Conference on Inventive Computation Technologies (ICICT), IEEE, 2022, pp. 171–176.
- P. Gavhane, D. Shinde, A. Lomte, N. Nattuva, and S. Mandhane, “Career path prediction using machine learning classification techniques,” Volume 8, Issue 3 of International Journal of Scientific Research in Computer Science, Engineering and Information Technology, 2021.